

Article

Machine Learning Models for Mapping Groundwater Pollution Risk: Advancing Water Security and Sustainable Development Goals in Georgia, USA

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Abstract: The widespread use of pesticides, such as atrazine and malathion, in agricultural systems raises significant concerns regarding the contamination of groundwater, which serves as a critical resource for drinking water. This study applies machine learning techniques to predict the concentrations of atrazine and malathion in groundwater across Georgia, USA, using 2019 data. A Random Forest classifier was employed to integrate various environmental and demographic factors, including pesticide application rates, precipitation, lithology, and population density, to predict pesticide contamination in groundwater. The models demonstrated high training accuracies of 100% and moderate average testing accuracy of 55% for atrazine and 60% for malathion across five iterations. The low test accuracy of the model, ranging from 50% to 75%, is likely due to overfitting, which can be attributed to the small dataset size and the complex nature of pesticide-contamination patterns, making it challenging for the model to generalize to unseen data. Feature importance analysis revealed that average pesticide usage emerged as the most influential factor for atrazine, while aquifer lithology and precipitation played crucial roles in both models. These results provide valuable insights into the dynamics of pesticide contamination, highlighting areas at greater risk of contamination. The findings underscore the importance of integrating environmental, geological, and agricultural variables for more effective groundwater management and sustainable agricultural practices, contributing to the protection of water resources and public health.



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1. Introduction

Pesticides are an integral component of modern agriculture, playing a pivotal role in protecting crops from pests, diseases, and invasive weeds [1,2]. Their widespread use has significantly enhanced food security by bolstering yields and mitigating post-harvest losses. Indeed, global pesticide usage now exceeds 5.6 billion pounds annually, with more than 1 billion pounds applied each year in the United States alone. Although such pervasive application is central to sustaining agricultural productivity, it also heightens concerns regarding health and environmental repercussions. In regions with limited resources and regulatory oversight, inadequate pesticide control measures often precipitate severe human exposure issues. An estimated 25 million agricultural workers worldwide suffer unintentional pesticide poisoning annually, reflecting stark inequalities in global agricultural safety [3]. Even in countries with tighter regulations, such as the United States,